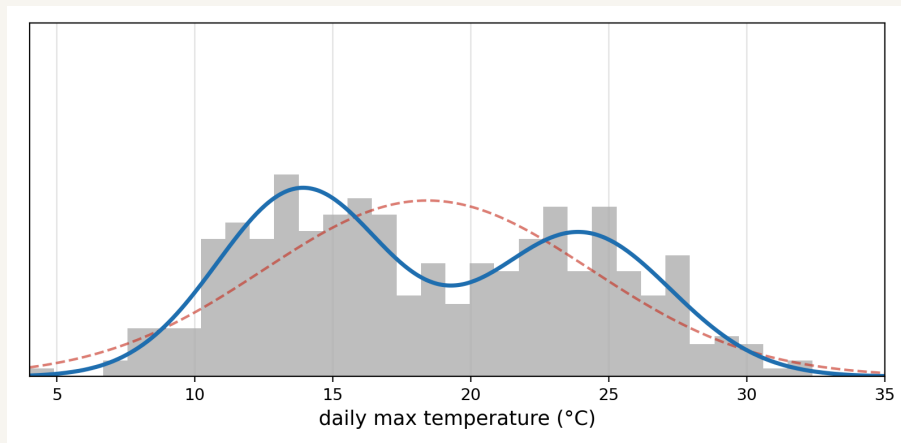


Gaussian mixture models

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When one Gaussian fails

A full year of temperatures



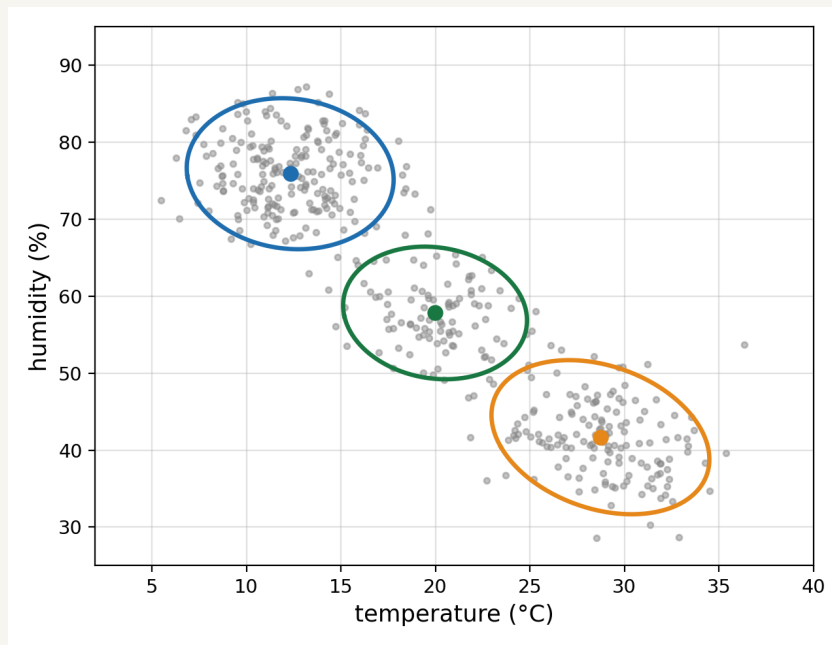
How can we summarize this kind of data ?

Level 1: the mean **Level 2:** $\pm$ the std **Level 3:** one Gaussian **Level 4:** several Gaussians

Is that a good model ?

Two Gaussians, **weighted and added** (a *mixture*) fit the year.

Same story in 2D



The mixture models the *population of clusters*.

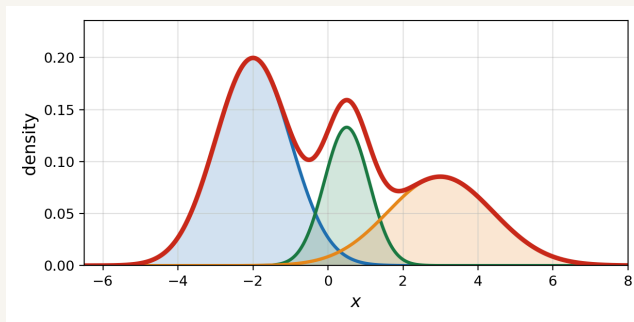
The mixture model

The mixture model

Definition: write $\mathcal{N}(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the Gaussian **density**. A **Gaussian mixture model** is a probability density given by a weighted sum of K Gaussian components:

$$f(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1$$

Parameters: the weights π_k , and one $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ per component.



$$\pi_1 = 0.5$$

$$\pi_2 = 0.2$$

$$\pi_3 = 0.3$$

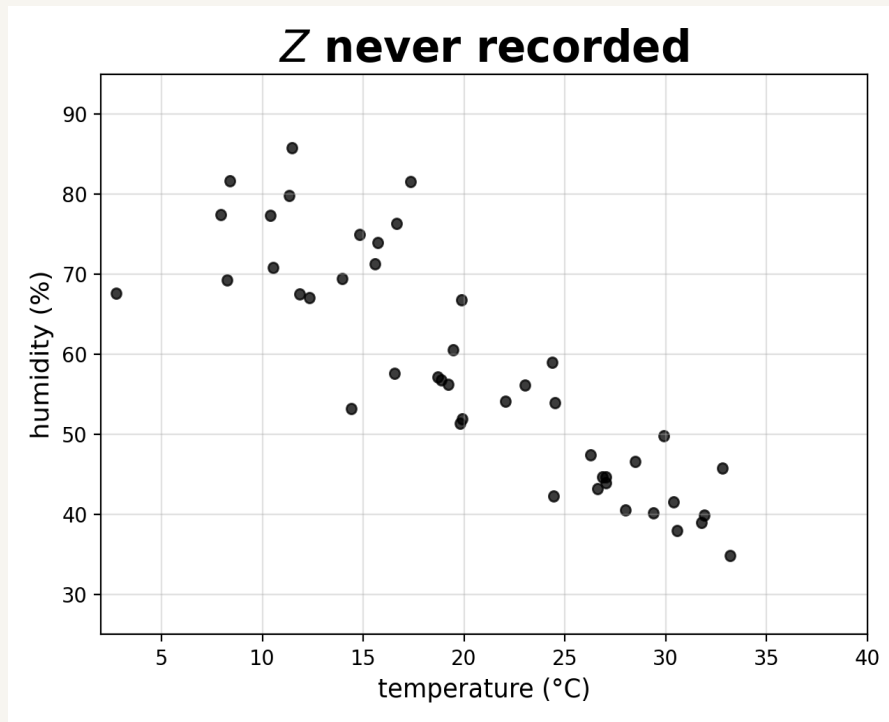
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The generative story

The model assumes an implicit generative process:

1. First randomly pick a component $Z \in \{1, \dots, K\}$ with $\mathbb{P}(Z = k) = \pi_k$
2. Then generate a data point $\sim \mathcal{N}(\mu_k, \Sigma_k)$

Z is **never observed** in practice: the generation is only *implicit*



Fitting a mixture

Exercise 1

Two likelihoods

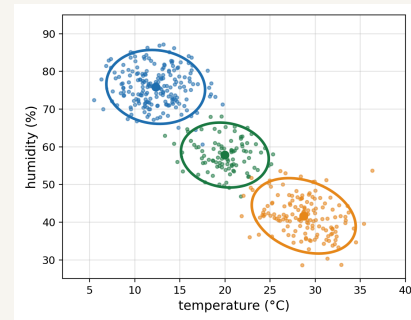
We observe $\mathbf{x}_1, \dots, \mathbf{x}_n$ and model them with a K -component GMM $(\pi_k, \mu_k, \Sigma_k)_{k \leq K}$.

1. The $\mathbf{x}_i$ are i.i.d. with density f . Write the log-likelihood $\ell(\theta)$ of the observed sample alone. What stops you from maximizing it term by term?
2. Let Z be the (unobserved) component, as $Z_i \in \{1, \dots, K\}$. Find the joint density of $(\mathbf{x}_i, Z_i = k)$ using Bayes theorem assuming $Z_i = k$.
3. Denote $\xi_{ik} = 1_{\{Z_i=k\}}$. Write the joint log-likelihood $\ell(\theta; (\mathbf{x}_i, Z_i)_{i=1..n})$ as a function of the ξ_{ik} and $\mathbf{x}_i$.
4. We do not observe the ξ_{ik} , so we integrate them out: compute $\mathbb{E}_{Z|X}(\ell(\theta; (\mathbf{x}_i, Z_i))_{i=1..n})$ using $\gamma_{ik} \stackrel{\text{def}}{=} \mathbb{P}[Z_i = k | x_i]$.
5. Assume the parameters $(\pi_k, \mu_k, \Sigma_k)_{k \leq K}$ are known, can you compute the γ_{ik} ?

A chicken-and-egg problem

If we **knew the probabilities** γ_{ik} : easy, optimization equivalent to single Gaussian:

$$\pi_k = \frac{n_k}{n}, \quad \mu_k = \frac{1}{n_k} \sum_i \gamma_{ik} \mathbf{x}_i$$
$$\Sigma_k = \frac{1}{n_k} \sum_i \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^\top$$



If we **knew the parameters**: we could compute the probabilities:

$$\gamma_{ik} = \mathbb{P}(Z_i = k \mid \mathbf{x}_i) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_i; \mu_j, \Sigma_j)} \quad (\text{Bayes})$$

The γ_{ik} are called **responsibilities**: how much component k is *responsible* for the point $\mathbf{x}_i$.

We know **neither**: the labels need the parameters, the parameters need the labels.

EM: Expectation–Maximization

EM alternates the two easy steps, using **soft** assignments γ_{ik} instead of hard labels.

Initialize π_k, μ_k, Σ_k , then repeat:

E step: soft-assign every point with the current parameters:

$$\gamma_{ik} \leftarrow \frac{\pi_k \mathcal{N}(\mathbf{x}_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_i; \mu_j, \Sigma_j)} \quad (\text{Probability that } \mathbf{x}_i \text{ belongs to cluster } k)$$

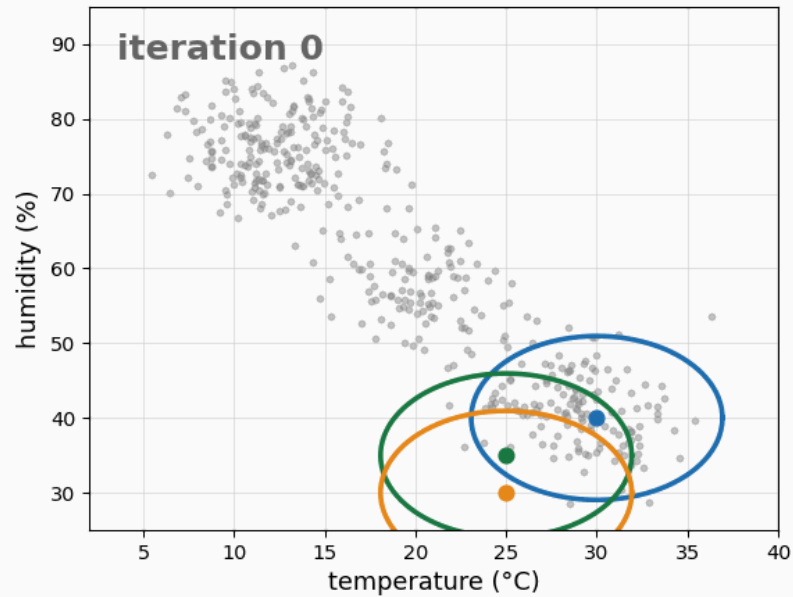
M step: refit each component on its softly-assigned points (with $n_k \stackrel{\text{def}}{=} \sum_i \gamma_{ik}$):

$$\pi_k \leftarrow \frac{n_k}{n}, \quad \mu_k \leftarrow \frac{1}{n_k} \sum_i \gamma_{ik} \mathbf{x}_i, \quad \Sigma_k \leftarrow \frac{1}{n_k} \sum_i \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^\top$$

(See section 2.3 for a detailed derivation).

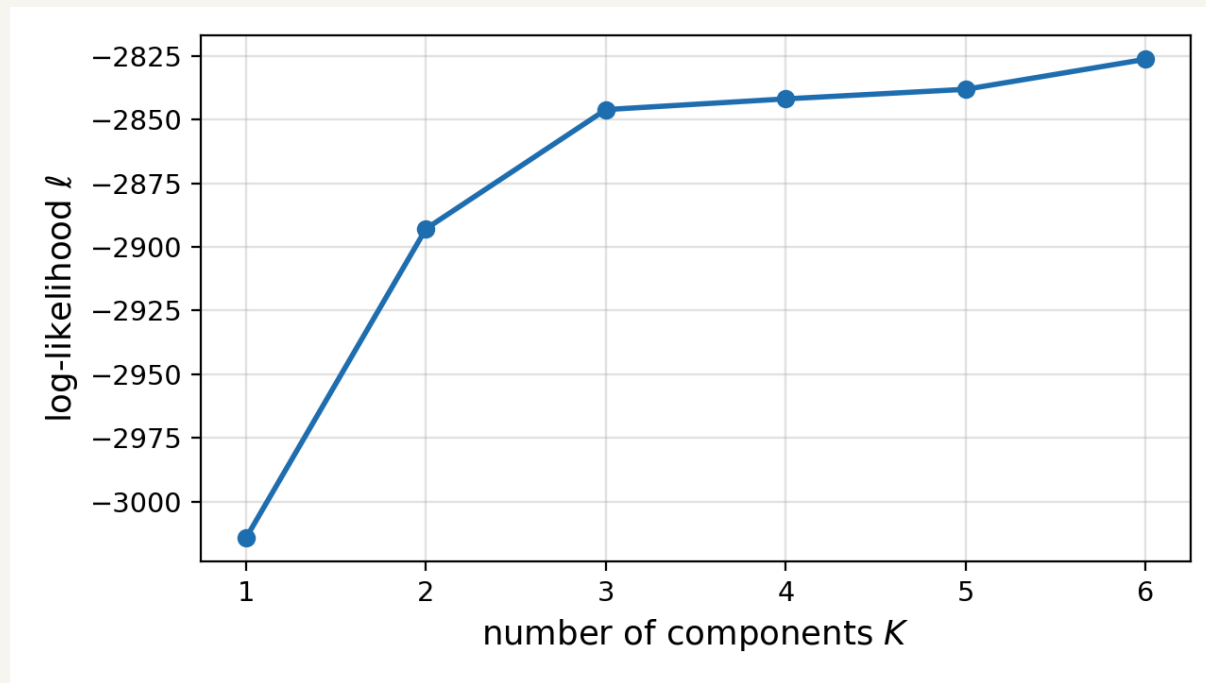
until the log-likelihood ℓ stops improving.

EM in action



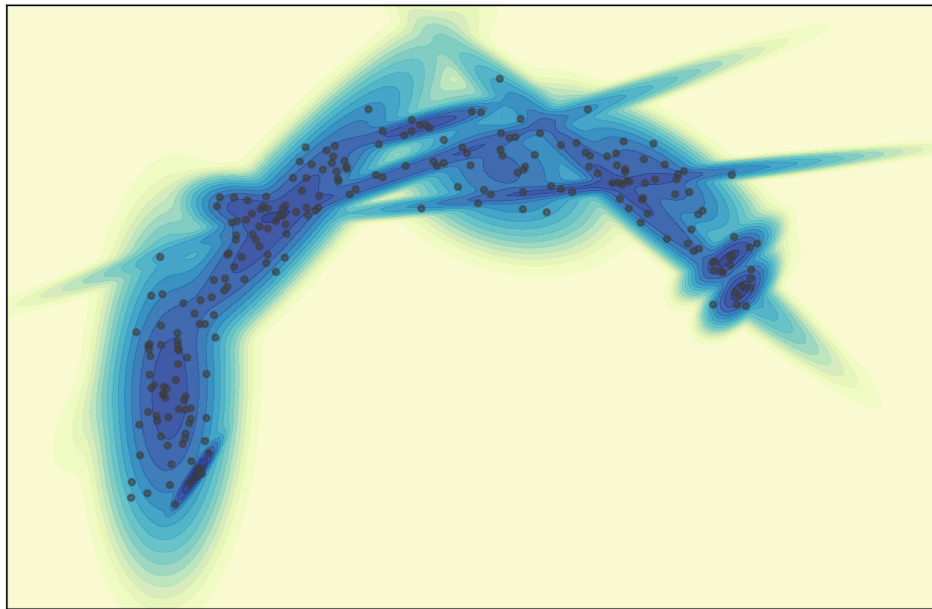
How many components ?

More components, better fit ?



ℓ **always** grows with K : should we just pick the largest K ?

Watching a model overfit



K = 1 K = 2 K = 4 K = 8 K = 12

A shape no small mixture fits well. What happens as K grows ?

Avoiding overfitting

Regularize the estimator: shrink each M-step covariance toward a multiple of the identity:

$$\Sigma_k \leftarrow (1 - \lambda) \widehat{\Sigma}_k + \lambda \bar{\sigma}_k^2 \mathbf{I}_d, \quad \bar{\sigma}_k^2 = \text{Tr}(\widehat{\Sigma}_k)/d$$

Simplify the model: restrict the covariance structure, trading expressiveness for parameter economy:

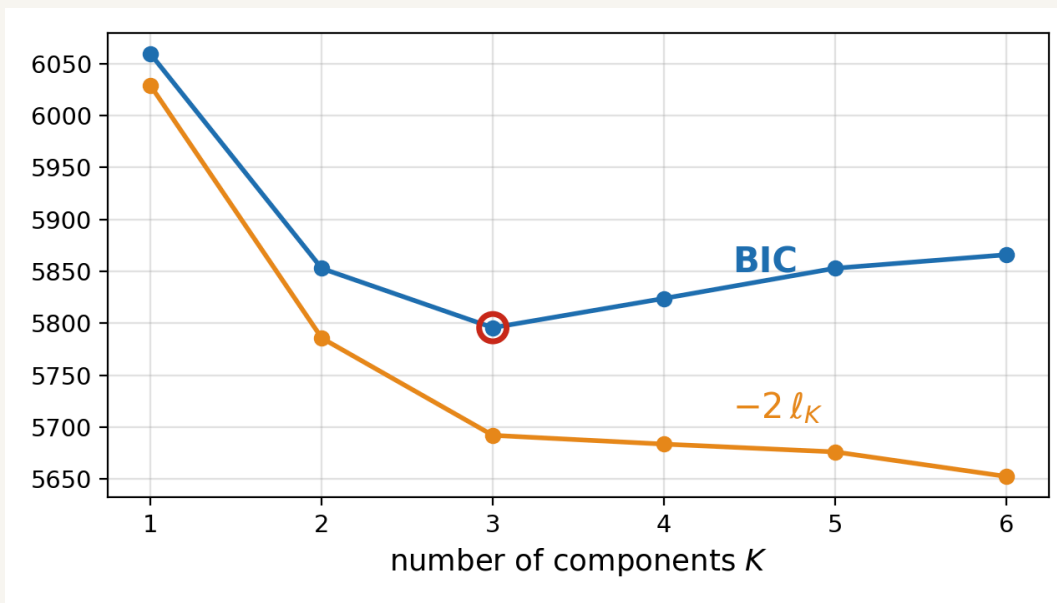
structure	constraint	covariance parameters
full	one free Σ_k per component	$K \cdot d(d+1)/2$
shared	$\Sigma_k = \Sigma$ for all k	$d(d+1)/2$
diagonal	no within-cluster correlation	$K \cdot d$
spherical	$\Sigma_k = \sigma_k^2 \mathbf{I}$	K

Reduce the dimension: do PCA to reduce the number of parameters to estimate.

Number of components K : maximize likelihood without inflating the number of parameters to be fit

Penalize complexity: BIC

$$\text{BIC}(K) = -2\ell_K + p_K \log n \quad p_K = \text{parameter count of the } K\text{-component model}$$

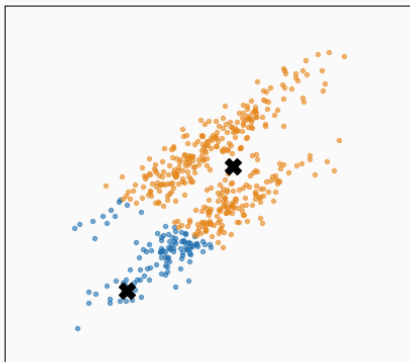


BIC trades fit against complexity: its minimum recovers the **true** $K = 3$.

GMM vs K-means

Same data, two algorithms

K-means vs GMM: how do you think its clusters will look ?



K-means



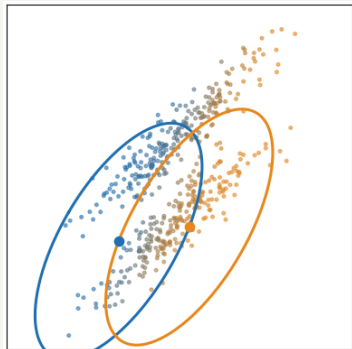
GMM

Can the two ever return **exactly** the same clusters ?

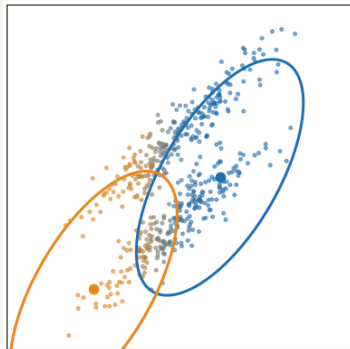
Yes, with spherical, equal-size clusters: $\Sigma_k = \sigma^2 \mathbf{I}$ shared, equal π_k .

Three starts, three answers

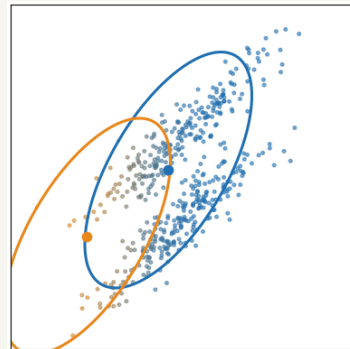
The same EM, the same data, three random initializations. The trails are the paths of the two means.



$\ell = -1733 \cdot 0.8\%$ misassigned



$\ell = -1916 \cdot 39\%$ misassigned



$\ell = -1733 \cdot 0.8\%$ misassigned



Warning: both algorithms depend on the **initialization**: a different starting point can give **different clusters**. In practice: we use multiple runs and pick the largest log-likelihood.

Compare the 1st and 3rd initializations, what do you notice ? **Orange and blue flipped !**

Contents

1. When one Gaussian fails
2. The mixture model
3. Fitting a mixture
4. How many components ?
5. GMM vs K-means